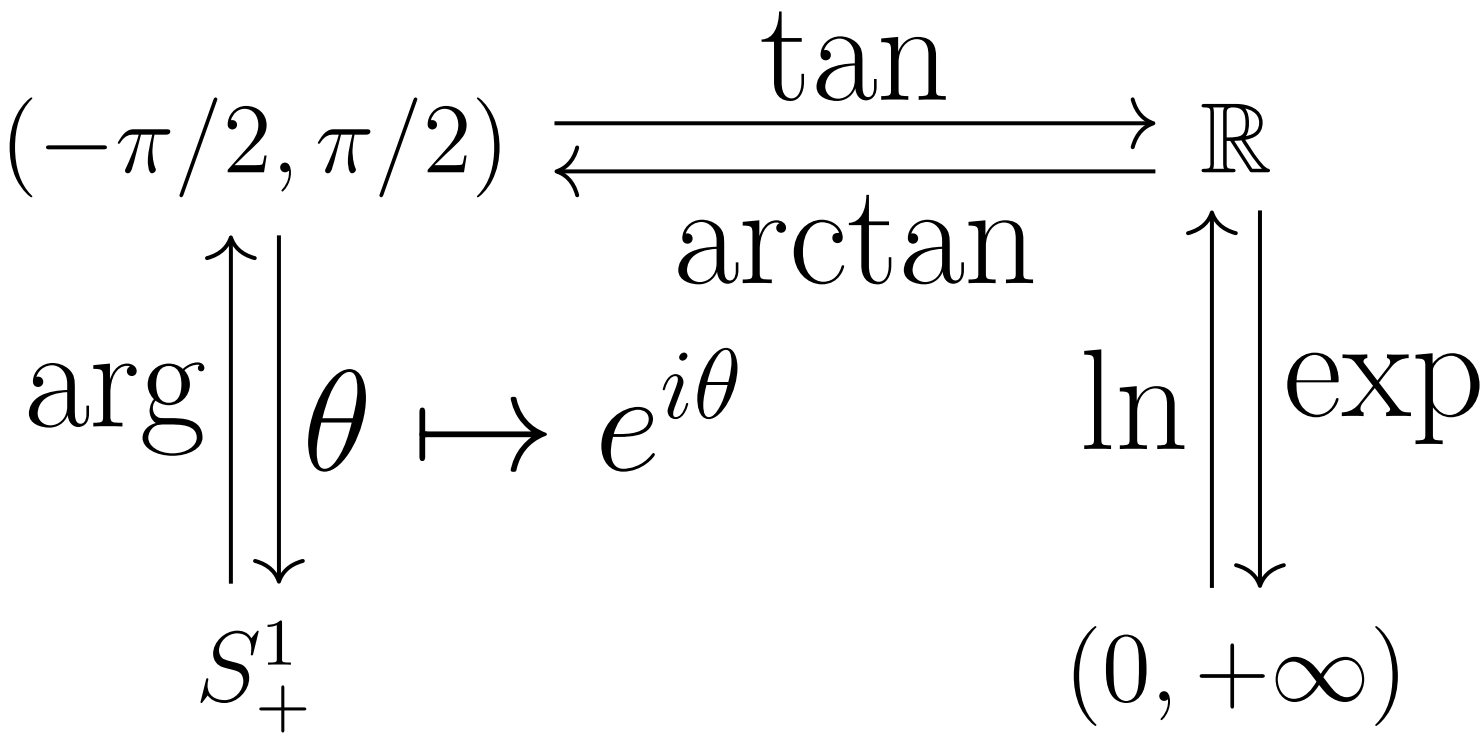
$(-\pi/2, \pi/2)$
$\tan$
$\arctan$
$\mathbb{R}$
$\arg$
$\theta \mapsto e^{i\theta}$
$\ln$
$\exp$
S^1_+
$(0, +\infty)$